

TS-TCN Component — Notes for Research Paper

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Methodology — TS-TCN Component

The Transaction-Sequence Temporal Convolutional Network (TS-TCN) is the temporal-modality detector of the platform. For each transaction under classification, it assembles the 32 most recent preceding transactions across the entire stream as a temporal context window, then classifies the transaction and identifies which preceding transaction most influenced the decision.

Each transaction is represented by 10 engineered features, all computable from standard mobile-money fields. The type-risk-weight feature is derived from the training partition only, to prevent data leakage. The window is built with a streaming buffer that snapshots the preceding transactions before appending the current one, guaranteeing causal validity.

Architecture

The model takes a 32×10 window and passes it through four dilated causal TCN blocks with dilation rates 1, 2, 4, and 8, followed by a self-attention layer (fraud_attention) and a dense classification head. Causal padding ensures no future-transaction information is used. The receptive field of 61 exceeds the window size of 32, so every position attends to the full predecessor history.

Layer	Configuration	Output
Input	32 transactions × 10 features	(32, 10)
TCN Block 1–4	Conv1D(96, k=3, causal), dilations 1/2/4/8	(32, 96)
fraud_attention	Self-attention, d_k=32	context + weights(32)
GlobalAvgPool1D	Temporal compression	(96)
Dense head	Dense(64,relu)+Dropout(0.3)+Dense(1,sig)	(1)

Total parameters: 218,945. Loss: BinaryFocalCrossentropy ($\gamma=2.0$). Optimizer: Adam ($lr=1e-3$).

Novel Contributions

- N1 — System-wide W=32 window: uses the recent transaction stream instead of per-account history, so the model works even for brand-new accounts with no prior activity.
- N2 — Dilated causal TCN (rates 1, 2, 4, 8): captures both immediate and long-range fraud escalation patterns, with causal padding guaranteeing no future-data leakage.
- N3 — fraud_attention layer: an intrinsic attribution mechanism that identifies the most influential preceding transaction with zero added latency, unlike post-hoc methods such as SHAP.
- N4 — Data-driven type risk weight: fraud prevalence per transaction type, computed from training data, making the encoding portable across deployment environments.

Feature Set

ID	Feature	Fraud Signal
F1	Drain ratio	→1.0 when account fully drained
F2	Log amount	Scale-normalised magnitude
F3	Post-transfer ratio	→0.0 when account emptied

ID	Feature	Fraud Signal
F4	Dest. was empty	Mule accounts start empty
F5	Dest. enrichment	Fraction of transfer that arrived
F6	Type risk weight	Data-driven prevalence by type
F7	Inv. dest ratio	→0.0 for large transfer to empty account
F8	Amt to orig	→1.0 when amount exhausts balance
F9	Hour of day	Fraud hour concentration
F10	Day of week	Weekly fraud patterns

Baseline Comparison

Three baselines were evaluated under an identical time-based split (train: step ≤ 595 ; test: step > 595 , 1,642 test fraud cases). The non-linear MLP baseline reaches high precision but limited recall, missing 680 fraud cases because it evaluates each transaction in isolation — the gap the TS-TCN targets with sequence context.

Model	F1	Recall	Precision	AUC
B0 — Rule (TRANSFER & amount>200K)	0.114	0.338	0.069	0.637
B1 — Logistic Regression	0.122	0.969	0.065	0.958
B2 — MLP (focal loss)	0.737	0.586	0.995	0.992
TS-TCN (proposed)	pending	pending	pending	pending

Output for Fusion Engine

The component returns a temporal risk score (0–1), a risk level (NORMAL / SUSPICIOUS / CRITICAL), the current-transaction evidence, and a triggering-predecessor attribution (the most influential preceding transaction with its feature vector and attention weight). The join key across all modules is `composite_id`, formatted as `nameOrig_step`. The output is delivered via POST `/api/v1/classify` within a 500 ms latency target.